Analysis of Station Fire Effects on Big Tujunga Creek Biotic and Abiotic systems, and a Road to Recovery

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Introduction:

Fire is a natural and sometimes necessary disturbance in many ecosystems. For a river ecosystem, fires can significantly affect the biotic and abiotic features like vegetation dynamics, soil properties, and hydrological processes. For instance, wildfires decrease canopy interception, water normally lost as evapotranspiration, and consumes ground cover, litter, duff, and debris(Moody & Martin 2001). And, they alter the chemical and physical properties of the soil which affects the river's hydrologic processes.

The Big Tujunga Creek, a large tributary of the Los Angeles River, is an important waterway within the San Gabriel Mountains of Southern California. The Big Tujunga Creek feeds into the Big Tujunga Wash and has a dam that prevents flooding and collects water for downstream communities. This creek serves as a critical resource, supporting diverse aquatic and riparian ecosystems, and it is characterized by its steep terrain and the willow riparian woodlands it runs through(Los Angeles County Public Work 2023). The creek is subject to dramatic wet and dry seasons, which, when combined with wildfires, can lead to significant changes in the river ecosystem. The creek and surrounding watershed have experienced numerous wildfires, with the Station Fire of 2009 being one of the most impactful in recent history.

The Station Fire, ignited on August 26, 2009, consumed an estimated 160,577 acres in the Angeles National Forest(NCEI 2009). The fire began during dry conditions in steep terrain covered with highly flammable, dry vegetation, and the area had been experiencing a severe drought. The wildfire claimed the lives of 2 firefighters, burned down homes and other infrastructure, and threatened natural and cultural resources, like watersheds(GAO 2011). Following the fire, communities downstream of the Big Tujunga Creek were at risk of increased flooding events. The absence of trees, shrubs, and other vegetation increases the speed and volume of runoff, and exposed soil is more vulnerable to erosion (NOAA 2023). Despite the evident impact of the Station Fire on Big Tujunga Creek, many questions remain regarding the long-term consequences of such fires on river systems.

There are many different studies that measure the impacts of wildfire on factors that regulate streamflow, water quality, sediment transport, and aquatic habitats(Rhoades 2019). But, the current knowledge of wildfire behavior and river ecosystem responses remains limited because of the limited amount of data taken before and after a wildfire event. Future research is needed to better understand the dynamics of wildfire on river ecosystems to best predict risks and how to best mitigate them, as extreme wildfires can jeopardize ecosystem health and water quality of freshwater sources. To address this gap in research, this study seeks to

answer the following question: how did the Station Fire of 2009 affect the hydrology and aquatic habitat conditions of Big Tujunga Creek? By using measurements from sites unburned and burned from the Station Fire, this study aims to contribute to a better understanding of river system responses to fire and propose future management practices to mitigate fire-related disturbances.

Methods:

1. Physical & Chemical Parameters

To assess the river quality, we examined chemical and physical parameters. The chemical parameters included acidity(pH), dissolved oxygen, phosphate, total nitrogen, electrical conductivity, temperature, and turbidity. There are ten physical parameters as listed in the Rapid Bioassessment Protocol which include epifaunal substrate, embeddedness, velocity/depth regimes, sediment deposition, channel flow status, channel alteration, frequency of riffles, bank stability (left and right bank), vegetative protection (left and right bank), and riparian vegetative zone width (left and right bank)(EPA Rapid Bioassessment Protocol, 2012). The Physical Habitat Quality Score is a sum of all the physical parameters scores, which scales the health of the stream compared to the "best attainable" situation.

2. Algae

Algal parameters were assessed as well to evaluate overall stream health, using Ash Free Dry Mass (AFDM) and Chlorophyll-*a* (Chl-a) tests, with 10 measurements at each site. AFDM is used to measure the abundance of algal biomass present in a sample, by burning any inorganic components present in the sample and then measuring the remaining weight of it (Moheimani, 2013). ADFM is relatively cost-effective though it has limitations: it does not distinguish between organic and inorganic matter, and algal and non-algal organic biomass, making it an underestimate of the total organic matter that is present. Chlorophyll-*a* is another useful method for measuring algal biomass using chlorophyll content. It is measured by obtaining a sample, extracting the chlorophyll pigment present in the sample, and then, the sample is passed through a spectrophotometer to measure how much light is absorbed by the algae. The higher the calculated absorbance, generally indicates that there are more algal cells present in the sample (Moheimani, 2013). Other factors such as the sample sizes and the temperature at which the algae is burned and weighed may also cause variability and uncertainty between samples (Liu, 2019). Chlorophyll-*a* is a more accurate method when measuring algal biomass samples, but it can have limitations due to the vast variability of chlorophyll content between algal species.

3. Macroinvertebrates

To collect samples of macroinvertebrates in the field a d-net was used by placing it directly on the substrate while the surrounding area was distributed either by kicking to loosen the substrate. Larger materials were then removed from the net so that remaining organisms could be analyzed (California Bioassessment protocol, 2003). After the sample was collected the Index of Biological

Integrity Score (I-IBI) could be calculated, using 7 common biotic metrics: coleoptera richness, EPT richness, predator richness, collector richness, intolerant species richness, non-insect richness, tolerant species richness. These scores were calculated using the raw data and table 3 from (Ode. et al 2005) to convert the raw data into a metric between 0-10, 0 being of lowest health and 10 being the highest. The SoCal I-IBI score allows us to determine the relative health of the river based on the richness of macroinvertebrates which make up the basis of the food chain and biodiversity.

4. Fish

In order to collect fish a 3-pass depletion electroshocking method was used to temporarily stun fish. Once stunned the fish that float to the surface and can then be counted to obtain abundance of the Santa Ana Sucker. One limitation of this method is that electroshockings efficacy depends heavily on both the size and species of fish, which can significantly alter abundance calculations (Peterson et. al 2004). After shocking some fish are then temporarily brought on board to determine a K factor, weight divided by length cubed (Nash et. al 2006). The K factor tells us the health of a fish in which a larger K value means a healthier fish.

5. Statistical tests:

A t-test was used to compare the difference in means of chemical parameters as well as invertebrate composition and fish density between burned and unburned sites. Post-hoc pairwise comparisons such as ANOVA were performed to test if means of physical parameters, algae, and fish condition were different among all sites, and a Tukey test allowed us to see specifically which sites showed significant statistical differences from one another. Analyses were performed using R packages “stats” and “agricolae”.

Results:

An ANOVA test showed significant differences on physical parameters including: substrate($p=1.65e-06$), embeddedness($p=7.57e-11$), velocity/depth regimes ($p=9.65e-10$), sediment deposition ($p=1.33e-09$), stability of left($p=9.97e-09$) and right banks ($p=1.47e-07$), vegetative protection of left($p=0.00124$) and right banks($p=9.34e-06$), riparian zone width on the left($p=2.03e-10$) and right($p=1.22e-10$), and the cumulative habitat scores($p=3.42e-14$). A Tukey test showed that all these parameters were higher in unburned sites, including the Physical Habitat Quality score represented in Figure 1. The Tukey test for sediment deposition distinguished unburned sites(U5, U6, U7, U8) from the burned sites(B1, B2, B3, B4), meaning there was significant difference between the unburned and burned sites but no significant variation within the unburned sites or within the burned sites. Additionally, flow and channel had p-values less than 0.05 from the ANOVA test, $p=0.193$ and $p=0.466$, respectively, so there was no Tukey test needed. And the frequency of riffles had a significant p-value($p=0.0469$), but the Tukey test revealed no significant difference between all of the sites.

For the chemical parameters, a two sample t-test was run to compare mean chemical parameters between the burned and unburned groups. For the acidity(pH), p=0.0005848 and t=7.8939; for phosphate, p= 3.273e-05 and t=12.378; for total nitrogen, p=0.001363 and t=5.7566; electrical conductivity had p=0.005256 and t=5.5073; and for turbidity, p=0.0005174 and t=-6.789. These parameters showed significant differences as the p-values were less than 0.05, and all parameters except for turbidity revealed higher means observed in the burned group than in the unburned. Lastly, temperature and dissolved oxygen showed no significant differences as the p-values were greater than 0.05(p=0.08041 and p=0.2136, respectively).

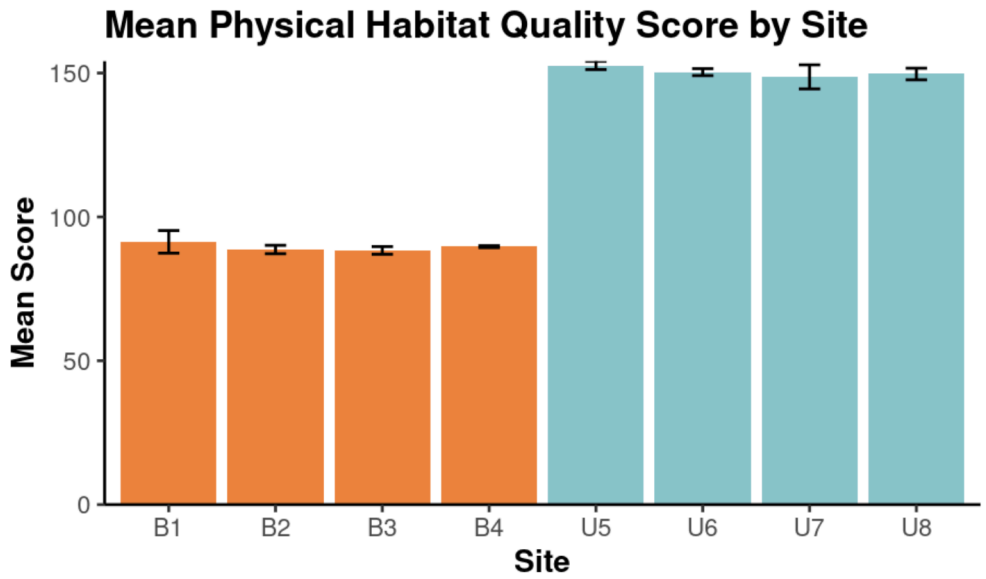

Fig. 1 Mean Physical Habitat Quality score by site. Unburned sites, U5, U6, U7, U8, have significantly higher mean physical habitat qualities scores than burned sites, B1, B2, B3, B4(Tukey test showed significant difference). Error bars represent standard error.

After performing a t-test on all 7 identified macroinvertebrate metrics, there appeared to be a significant difference in the means among the majority of metrics between burned and unburned sites. Coleopter, EPT, predator, collector, tolerant, all showed statistical significance with p-values in order of listed metric, (<0.001, 0.0015, <0.001, <0.001, 0.003) and t-statistics in order of listed metric, (-14.142 , -6.6398, -9.8198, 10.961, 5.8368). This test was also performed on the SoCal B-IBI scores and showed significant difference between treatment groups (*Fig. 2*, p-value <0.001, t-statistic = -16.933). The difference in fish density between burned and unburned sites also showed a significant difference in means, with a p-value of 0.015, and t-statistic of -3.3828. After running an ANOVA on fish body condition there was also a difference detected among sites with a p-value of <0.001, with the largest difference as determined by a Tukey test between burned and unburned sites.

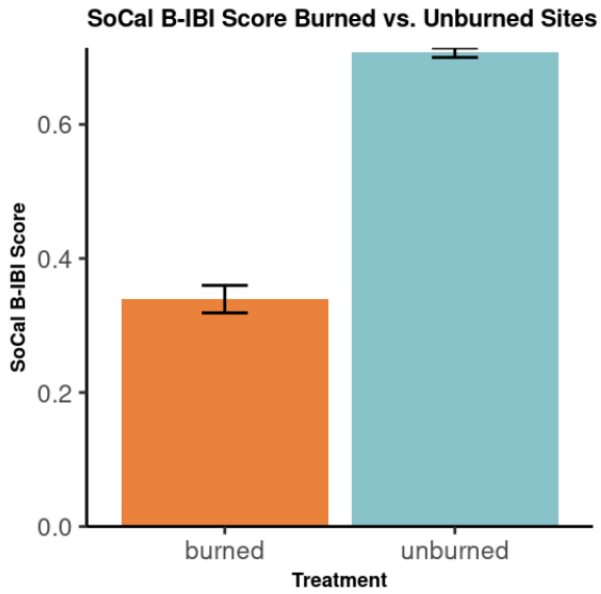

Fig. 2 SoCal B-IBI Scores from Burned vs. Unburned Sites. SoCal B-IBI scores were significantly higher in sites that were not burned in the Station Fire (t-statistic -16.933, $p < 0.001$) n = 4 for each treatment group. Error bars represent standard error of the mean.

Upon running an ANOVA and Tukey test for each of the algal parameters AFDM and Chl-*a*, it was found that on average the burned sites had a higher Chlorophyll-*a* measurement, and higher measurements of Ash Free Dry Mass over the unburned sites. A T-test showed that these measurements were statistically different (indicated by $P << 0.05$) at each of the sites. AFDM's t-test yielded a p-value ($P = 0.000239$), indicating a statistical difference in the given metric for the unburned and burned sites. The t-test measured for the Chl-*a* metric indicated statistically different results as well, with a p-value of 0.000079. These results indicated an increase in algal biomass on the burned sites over the unburned sites.

The Tukey test for the metric of AFDM showed statistical similarities between sites B2, B4, B3, U6 and U5. Site B2 was statistically different from sites U8 and U7. Site U8 showed statistical similarity with sites B2, B4, B3, U6 and U5. Lastly sites B4, B3, U6 and U5 appeared to show the most statistical similarity out of the rest of the other sites. The Tukey test for the Chl-*a* metric indicated statistical difference between site B3, and sites U5, U6, U7, and U8, and statistical similarity between site B3, and sites B1, B4, and B2. Lastly, there is statistical similarity between all groups.

Discussion:

Chemical and Physical Parameters:

Our results indicate significant changes in chemical and physical parameters, aligning with broader environmental trends and previous studies. Nitrogen was observed to be greater in burned sites than unburned likely because of increased ash and soil runoff, loss of vegetation,

and release of nitrogen from soil. A previous study found that burning increases the availability of nutrients, like nitrogen as organic matter is combusted through fire(Debano 1991). Additionally, pH of the river in burned sites was significantly greater than the unburned sites which aligns with what other studies have found. Fire has been found to increase pH as ash formation on soil releases alkaline elements(Memoli 2020). As for physical parameters, the unburned sites had significantly greater embeddedness and velocity regimes. As mentioned, fires consume organic matter in soils, and reductions in the interception and infiltration of rain in burnt watersheds results in greater overland flow, which is exacerbated by increased surface velocities and connectivity of flow paths due to bare ground(Moody 2013). So, lower embeddedness deposition after a fire can be attributed to increased flow.

Algae:

The results and figures indicated a significant increase in biomass when measured with Chl-a and AFDM methods, as seen in *Fig.3* and *Fig.4*. Since the Chl-*a* method measures total biomass and abundance of algae in a given site, an increase in nitrogen and other nutrients deposited into the water column from the wildfire was the likely cause of algal growth at each of the burned sites. The fire could have increased sedimentation of dissolved organic matter, which also could have contributed to the increase in algae(Cooper, 2014). The unburned sites likely did not accumulate this increase in nitrates and nutrients from the fire, and the algal biomass was not affected. Ash Free Dry Mass on average appeared to be significantly higher in the burnt sites as well. This could be also attributed to the increase in algal growth, since the AFDM metric represents the inorganic ash subtracted from the organic algal biomass. These results are consistent with the increase in Chlorophyll-*a* content, measured and seen in *Fig. 3*.

These results align with results measured from previous studies. When studying the effects of wildfires on the stream algal communities in Southern California, the Santa Ynez mountains in Santa Barbara County were amongst the sites studied, representing steep coastal mountains with Mediterranean climate, like the Big Tujunga. Researchers found that the macroalgal mass, and Chlorophyll-*a* measured was 5 to 20 times higher in the burned sites with riparian vegetation (Klose, 2015). This was due to the fire burning off the riparian vegetation, causing an increase in runoff and erosion, thus increasing nutrient deposition into the river causing eutrophication, and altering soil composition and affecting water quality (Klose, 2015). This pattern can also be observed in different regions. A study measuring the effects of fires on the subarctic lakes of northern Alberta, showed a more than 2 fold increase in dissolved phosphorus, and at least a 1.2 fold increase in Nitrogen concentrations for the burnt catchment areas (McEachern, 2000). Such nutrient enrichment will cause a rapid increase in algal abundance, which can change the ecosystem and cause macrophyte communities to be replaced by algal blooms (Pederson, 1996).

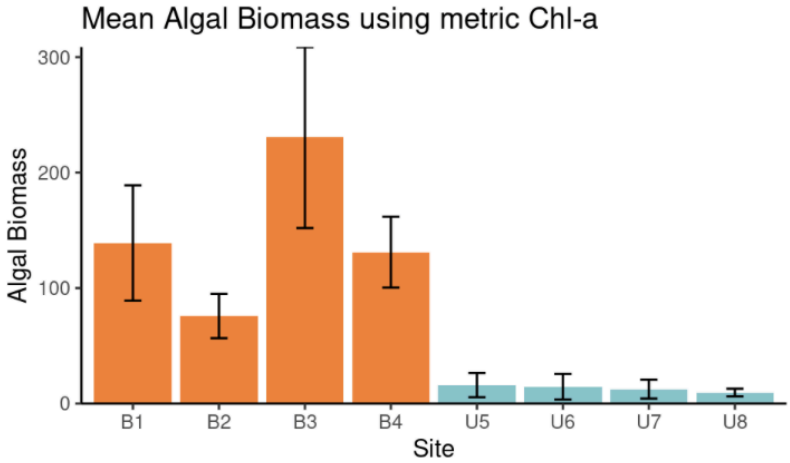

Fig. 3 Mean Algal Biomass Measured by metric Chl-*a*. B1, B2, B3, and B4 represent the burned sites, while U5, U6, U7 and U8 represent the unburned sites. On average, algal biomass is higher in the burned sites over the unburned sites. Average Chl-*a* measurement for the burned sites = 144.06 mg/L. Average Chl-*a* measurement for unburned sites = 13.15 mg/L.

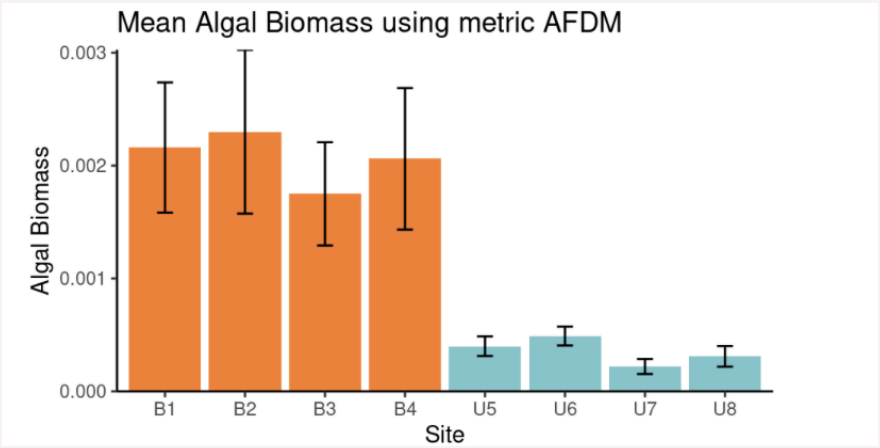

Fig. 4 Mean Algal Biomass Measured by metric AFDM. B1, B2, B3, and B4 represent the burned sites, while U5, U6, U7 and U8 represent the unburned sites. On average, algal biomass is higher in the burned sites over the unburned sites. Average AFDM measurement for the burned sites = 0.00207 mg/L. Average Chl-*a* measurement for unburned sites = 0.00035 mg/L.

Invertebrate:

The results show that there was a significant difference in predator richness between burned and unburned sites. This was likely due to the fire increasing overland water flow as there was less vegetation to transpire/intercept water. These flows can greatly alter habitat as well as food availability and quality (Erdozain et. al, 2024). EPT richness was also significantly lower in burned areas showing a cumulative decline in aquatic insect density and diversity. Post-fire sediment is likely to increase the transport of pollutants such as metals and hydrocarbons which can directly harm much of the invertebrate biota, and alterations in light availability from lack of vegetation can also alter primary production hindering the growth of algae which is their primary food source(Erdozain et. al, 2024). Our results are consistent

with other studies including those conducted in Yellowstone National Park in which there were declines in macroinvertebrate abundance, richness, and diversity(Lawrence and Minshall, 1994). However, results differ significantly based on surrounding ecology and both severity and duration of burn (Gresswell, 1999). In areas burned more severely the indirect effect on macroinvertebrates comes from more extreme runoff and sediment deposition (Minshall et al. 1997)

Fish:
Changes in fish density and body condition are caused by both direct and indirect effects of the Station Fire. Density of fish decreasing among burned sites is likely due to direct mortality caused by changes in the chemical composition of the water from smoke. This would be in line with other studies in which mortality increased due to pH change after only 30 minutes of exposure (Cushing and Olson, 1963). In Yellowstone National Park, while there was no observed change in growth or abundance of fish in the river 1 year post-fire, there was an immediate decline in density(Jones et al. 1993). Additionally, with changes in body condition for fish that did not suffer immediately, mortality would decrease again due to pH changes as well as food availability decrease from lack of allochthonous and autochthonous input and decrease in invertebrates which make up the basis of the consumer food chain. Current evidence suggests that in many cases fish are resilient to even the most severe burns and long term detrimental effects only occur in highly isolated populations where anthropogenic affect is high (Gresswell, 1999).

Synthesis and Overall Conclusions:
The major effects of the fire on the Big Tujunga Creek ecosystem were substantial decreases in habitat quality, algal biomass, invertebrate density and fish density/condition. These changes were due to the burn and its cascading effects including altering the pH of the water, substrate deposition and bank stability which in turn negatively impacted many biotic systems within the river. These findings show similar results to previous studies in which there were significant immediate effects on rivers post fire both biotic and abiotic, heavily dependent on burn severity and intensity (Gresswell, 1999). Overall, while algal and invertebrate communities are likely to show recovery within a few years after a fire, without any additional mitigation tactics, fish populations are heavily dependent on existing ecological networks and habitat conditions and may take longer to recover; as previously mentioned, isolated species are more likely to suffer long term consequences(Gresswell, 1999). Regular monitoring and management are essential to facilitate recovery and support ecosystem resilience after fire events. Studies that record the same environment, annually post-fire could be used to test the predictions of algal, invertebrate and fish communities, as well as annual or monthly water samples to test for chemical parameters, using unburned sites as the control.

Mitigation Plan:
Alternative studies show that wildfires can have dramatic impacts on streams in the short term, but most biological factors recover within 5-10 years naturally (Erdozain et. al, 2024). However, this is not an ideal mitigation strategy, especially in an area such as Los Angeles where the increasing effects of climate change are only making wildfire more prevalent. There are ways in which we can hasten the recovery of burned sites to pre-fire conditions by using methods such as flood plain rewetting and re-planting of native and diverse vegetation (Erdozain et. al, 2024). We know that certain activities such as damming and channelization decrease a river's connection to ground water and thus make the surrounding area more vulnerable to wildfire. A restoration technique such as floodplain connectivity would lead to groundwater replenishment and increased hyporheic exchange by restoring historical braided channel forms and physical structure of the river previous to anthropogenic interception (Pugh et. al, 2022). This would increase the soil moisture, decreasing future burn severity, and lead to a higher recovery and resilience of vegetation and soil microbiome (Pugh et. al, 2022). Floodplain rewetting while predictively effective would require substantial monetary input and labor. Removing one or both of the dams along the Big Tujunga creek while costly would also positively influence fish populations post fire as they would be less isolated, leading to better reproduction and recruitment rates and nutrients would be better distributed throughout the river. Fires typically immediately increase the nitrogen and phosphorus levels in streams, while this is typically uptaken by biota to mitigate the effects of harmful algal blooms (Gresswell, 1999), dams create an uneven distribution of sediment which could exceed the level to which invertebrate populations, especially diminished ones in burned areas are able to effectively uptake the increase in nutrients. Thus removing the dams could have significant cascading effects to prevent future fires and restore the watershed post fire. Additionally, replanting vegetation around the banks of the river would help return to pre-burn levels of sediment deposition through increased interception as well as restore the chemical composition of runoff entering the river through transpiration. Vegetation would also increase canopy cover helping prevent photoinhibition and increase allochthonous input to support a bottom up food chain, increasing the nutrient supply for invertebrates and fish. Furthermore, ensuring that the replanted vegetation is both native and diverse will lead to a higher resilience and quicker recovery in the event of proceeding fires (Pugh et.al, 2022). While not encompassing the full scope of recovery these are a few methods which could significantly contribute to the restoration of burned sites in Big Tujunga Creek.

Broader implications
The Thomas fire occurred in Ventura county in the Los Padres National Forest on December 4th, 2017. It was a result of an arcing power line, and it destroyed 1,063 structures, burned 281,893 total acres, and spread over 42 miles from Santa Barbara, to Fillmore. An important factor that helped reduce the amount of homes and acres destroyed by the Thomas fire, were

the pre-fire mitigation strategies that the community surrounding the Los Padres National Forest had been implementing for 2 years prior to the fire (Gabbart, 2019). Since the fire, there have been attempts at the restoration of the area, similar to the ideas proposed in our mitigation plan about replanting native and diverse vegetation. The pre-fire mitigation consisted of planting native vegetation that is adapted to fire to cover the preserves. As of now, native shoots of Laurel, Yuca and Oak trees have been observed to be making a return to the area, and the soil is starting to be filled with native seed banks that are starting to germinate. The Ojai Valley Land Conservancy has been making efforts to help the restoration of the ecosystem, by weeding non-native plants to aid the recovery and growth of the native plants. They have had volunteer events to clean up the burned irrigation systems, and are planning to replant 6 acres of the site with coast live oaks and native shrubs (Squarespace, 2023).

A study was conducted on the effects of the Thomas fire on the dynamics of the Ventura river. Between 1971 and 2009, the fine sediment concentrations had decreased overall in the Venture river. It was found that in 2017, the fire caused a severe increase of at least 200 fold in runoff and fine and suspended sediment flux. This indicated concentrations of sediment that were much higher than any predicted concentrations, causing potential impairment to the ecosystems in the Ventura river (Jumps, 2022). Given this data, the parameters we used for the study in the Tujunga Creek could be used to assess the stream health of the Ventura river. Sediment deposition is a physical characteristic that can be assessed with statistical analysis (ANOVA and Tukey tests), and can be incorporated to address the stream's overall health. Other parameters could be relevant as well, such as chlorophyll-a and AFDM, since the study indicates an increase in runoff and erosion, seeing an increase in chlorophyll-a and ash free dry mass could be another indicator of impaired stream health after the fire.

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